

Ryan Cho

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-stack engineer with 7+ years delivering scalable APIs, data-intensive web applications, and AI-enabled workflows across financial services. Expertise in Python, FastAPI, TypeScript, React, and Node.js with production experience integrating LLM services, RAG pipelines, and real-time analytics dashboards. Proven track record building reliable platforms from prototype to production in fast-paced, cross-functional environments.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and delivered a **full-stack portfolio analytics platform** using React, TypeScript, and Python FastAPI serving 200+ internal users with sub-second query response across multi-source financial datasets.
- Built **scalable REST and WebSocket APIs** on Node.js and FastAPI to power real-time position monitoring, reducing manual reconciliation effort by 40% for operations teams.
- Designed a **RAG-based document retrieval service** integrating LangChain and vector embeddings to surface relevant fund documentation, cutting analyst research time on recurring queries.
- Led **production deployment automation** with Docker, AWS ECS, and CI/CD pipelines that improved release cadence from biweekly to twice weekly while maintaining 99.9% uptime.
- Implemented an **interactive reasoning dashboard** in React with streaming LLM outputs, enabling portfolio managers to trace model conclusions and validate recommendations in real time.
- Owned the **PostgreSQL and Redis data layer** for high-throughput financial ingestion pipelines processing 500K+ daily market events with idempotent async workers.
- Partnered with **quantitative researchers and product leads** to productionize prototype ML scoring models into governed API endpoints with versioning and audit logging.
- Established **monitoring and performance standards** using Datadog and structured logging, reducing mean incident detection time from hours to under 15 minutes.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Java and Python microservices** supporting trade lifecycle workflows across fixed income desks, handling 10K+ daily transactions with consistent SLA adherence.
- Built **internal React dashboards** for risk exposure visualization consumed by 50+ technology and front-office stakeholders across three regional teams.
- Integrated **Bloomberg and internal market data APIs** into normalized ingestion pipelines with validation rules that reduced downstream data quality incidents by 30%.
- Implemented **async batch processing jobs** for end-of-day reconciliation reports, cutting overnight processing windows by 25% through query optimization.
- Contributed to **API gateway standardization** efforts, documenting REST contracts and pagination patterns adopted by five downstream application teams.
- Supported **relational database migrations** on PostgreSQL and Oracle schemas for regulatory reporting modules with zero unplanned downtime during cutover windows.
- Collaborated in **Agile delivery cycles** with cross-functional technology and compliance partners to ship prioritized enhancements across two major release trains annually.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **full-stack features** for restaurant operations SaaS using React and Node.js, improving order tracking visibility for pilot merchant locations.
- Developed **RESTful API endpoints** with Express and MongoDB for menu and inventory modules used across web and mobile client integrations.
- Wrote **unit and integration tests** with Jest and Supertest that increased coverage on critical payment flows and reduced regression defects in QA cycles.
- Participated in **code reviews and sprint planning** with senior engineers, shipping four customer-facing enhancements during the internship tenure.
- Optimized **frontend rendering performance** on data-heavy admin screens through memoization and pagination, improving page load times on low-bandwidth connections.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Delivered a **responsive nonprofit website** with HTML, CSS, and JavaScript, increasing mobile traffic engagement for fundraising campaign pages.
- Implemented **content management workflows** enabling non-technical staff to publish updates without developer intervention.
- Configured **analytics and SEO metadata** to improve discoverability of foundation programs and event listings.
- Collaborated with **stakeholders** to iterate on layout and accessibility improvements ahead of public launch deadlines.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML5, CSS3, Next.js, Redux, WebSockets, D3.js

Backend: Python, FastAPI, Node.js, Express, REST APIs, GraphQL, async programming, LangChain

Databases: PostgreSQL, MongoDB, Redis, Oracle, vector databases, pgvector

Cloud & DevOps: AWS, Docker, Kubernetes, ECS, CI/CD, Terraform, GitHub Actions

Monitoring & Tools: Datadog, Prometheus, Grafana, Jest, Pytest, profiling, structured logging

Workflow & PM: Agile, Scrum, Jira, cross-functional collaboration, technical documentation